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COMMUNICATION DEVICE

Abstract

A communication device includes an antenna; a first generation unit; a duplexer; a second generation unit; a suppression unit; a first detection unit; and an adjustment unit. The duplexer receives the carrier wave generated by the first generation unit from an input terminal and outputs the carrier wave to the antenna from an input and output terminal, and outputs a signal input from the antenna to the input and output terminal from an output terminal. The second generation unit generates a cancel signal by changing an amplitude and a phase of the carrier wave. The suppression unit suppresses a self-interference signal in an output signal from the output terminal by using the cancel signal. The first detection unit detects a mismatching state of impedance matching related to the antenna. The adjustment unit adjusts, in response to detection of the mismatching state, an antenna impedance to eliminate the mismatching state.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-031187, filed on Mar. 1, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate to a communication device, a communication method, and an RFID reading device.

BACKGROUND

[0003] In a communication device that shares an antenna for transmission and reception, a part of a transmitted signal may be superimposed on a received signal and flow into a reception system. A transmitted signal component superimposed on the received signal becomes a self-interference signal, and there is a concern that communication quality may be degraded due to saturation of the reception system and an increase in noise.

[0004] Therefore, a technique is known in which a cancel signal having a phase opposite to that of the self-interference signal is generated based on the transmitted signal, and the self-interference signal is cancelled using the cancel signal.

[0005] The self-interference signal varies depending on a surrounding environment. Therefore, before receiving a response signal from a wireless tag, a carrier wave is transmitted to generate a self-interference signal, and a cancel signal is adjusted so as to effectively cancel the self-interference signal.

[0006] However, variations in an antenna impedance due to changes in the surrounding environment cause variations in the self-interference signal, and the self-interference signal may not be appropriately cancelled by the adjusted cancel signal.

Description

DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a block diagram showing a circuit configuration of a reading device according to an embodiment;

[0008] FIG. 2 is a block diagram showing a circuit configuration of main parts of a matching adjustment unit;

[0009] FIG. 3 is an immittance chart showing adjustment amounts of a plurality of matching circuits;

[0010] FIG. 4 is a flowchart of controlling;

[0011] FIG. 5 is a flowchart of controlling;

[0012] FIG. 6 is a diagram showing an actual measurement example of variations in a phase of a reflected signal over time; and

[0013] FIG. 7 is a diagram showing an actual measurement example of a change in cancellation amount due to a phase error.

DETAILED DESCRIPTION

[0014] Under the circumstances described above, it is desired to appropriately cancel the self-interference signal even if the antenna impedance varies. Therefore, in general, according to one embodiment, a communication device that is capable of appropriately cancelling a self-interference signal even if an antenna impedance varies is provided.

[0015] According to one embodiment, a communication device for receiving a response wave emitted from a wireless tag based on a received carrier wave includes an antenna, a first generation

unit, a duplexer, a second generation unit, a suppression unit, a first detection unit, and an adjustment unit. The first generation unit generates a carrier wave. The duplexer receives the carrier wave generated by the first generation unit from an input terminal and outputs the carrier wave to the antenna from an input and output terminal, and outputs a signal input from the antenna to the input and output terminal from an output terminal. The second generation unit generates a cancel signal by changing an amplitude and a phase of the carrier wave generated by the first generation unit. The suppression unit suppresses a self-interference signal in an output signal from the output terminal by using the cancel signal generated by the second generation unit. The first detection unit detects that impedance matching related to the antenna is in a predetermined mismatching state. The adjustment unit adjusts, in response to detection of the mismatching state by the first detection unit, an antenna impedance to eliminate the mismatching state.

[0016] Hereinafter, an embodiment will be described with reference to the drawings. In the following description, a reading device that reads data stored in a radio frequency identification (RFID) tag will be described as an example. The reading device performs wireless communication with the RFID tag when reading the data, and is an example of a communication device. The RFID tag is an example of a wireless tag.

[0017] FIG. 1 is a block diagram showing a circuit configuration of a reading device **100** according to the embodiment.

[0018] The reading device **100** includes an oscillator **11**, a phase shifter **12**, a digital to analog (DA) converter **13**, a quadrature modulator **14**, a band-pass filter (BPF) **15**, a power amplifier **16**, a low-pass filter (LPF) **17**, an antenna duplexer **18**, a matching adjustment unit **19**, a feed line **20**, an antenna **21**, a variable attenuator **22**, a variable phase shifter **23**, a DA converter **24**, a power combiner **25**, a quadrature detector **26**, an LPF **27**, an alternating current (AC) coupling amplifier **28**, an analog to digital (AD) converter **29**, an LPF **30**, an AD converter **31**, a control unit **32**, and a memory **33**. The control unit **32** includes a central processing unit (CPU) **321** and a field programmable gate array (FPGA) **322**. The antenna **21**, or the feed line **20** and the antenna **21** may not be in the reading device **100**, and any separate device may be connectable.

[0019] The oscillator **11** generates a sine wave having a predetermined frequency as a carrier wave.

[0020] The phase shifter **12** shifts a phase of the carrier wave generated by the oscillator **11** by 90 degrees, and outputs a cosine wave as another carrier wave.

[0021] The oscillator **11** and the phase shifter **12** are examples of a first generation unit.

[0022] The DA converter **13** converts two-system transmitted baseband signals output in a digital state from the CPU **321** into analog signals. Hereinafter, the two-system transmitted baseband signals are referred to as an I signal and a Q signal, respectively.

[0023] The quadrature modulator **14** receives the I signal and the Q signal converted into analog signals by the DA converter **13** as modulated waves. The quadrature modulator **14** receives the carrier wave generated by the oscillator **11** and the carrier wave output from the phase shifter **12** as carrier waves of an I system and a Q system, respectively. The quadrature modulator **14** obtains a transmitted signal by quadrature modulation.

[0024] The BPF **15** removes a low-frequency component and a high-frequency component from the transmitted signal obtained by the quadrature modulator **14** for band limitation.

[0025] The power amplifier **16** amplifies power of the transmitted signal that passes through the BPF **15** to a level suitable for wireless transmission.

[0026] The LPF **17** removes a harmonic component from the transmitted signal amplified by the power amplifier **16**.

[0027] The transmitted signal becomes a signal for wireless transmission by processing in the BPF **15**, the power amplifier **16**, and the LPF **17**. That is, the BPF **15**, the power amplifier **16**, and the LPF **17** generate a transmitted signal for wireless transmission.

[0028] The antenna duplexer **18** includes an input terminal TI, an input and output terminal TIO, an output terminal TOA, and an output terminal TOB. The transmitted signal that passes through the

LPF **17** is input to the input terminal TI. The antenna duplexer **18** outputs the transmitted signal input to the input terminal TI from the input and output terminal TIO and the output terminal TOB. The antenna duplexer **18** outputs a signal input to the input and output terminal TIO from the output terminal TOA. A signal output from the output terminal TOA of the antenna duplexer **18** is a signal obtained by combining a received signal generated in the antenna **21** and a self-interference signal to be described later.

[0029] Hereinafter the signal will be simply referred to as a received signal. The antenna duplexer **18** is an example of a duplexer.

[0030] The matching adjustment unit **19** matches an impedance of the antenna **21** within a predetermined control target range.

[0031] The feed line **20** supplies a transmitted signal output from the input and output terminal TIO of the antenna duplexer **18** to the antenna **21**. The feed line **20** transmits a received signal generated in the antenna **21** to the input and output terminal TIO of the antenna duplexer **18**.

[0032] The antenna **21** emits a radio wave corresponding to the transmitted signal supplied from the feed line **20**. The antenna **21** generates an electrical signal corresponding to the incoming radio wave as a received signal.

[0033] The variable attenuator **22** attenuates a transmitted signal output from the output terminal TOB of the antenna duplexer **18** with a gain corresponding to a gain setting signal supplied from the DA converter **24**.

[0034] The variable phase shifter **23** changes a phase of the transmitted signal attenuated by the variable attenuator **22** by a phase shift amount corresponding to a phase shift amount setting signal supplied from the DA converter **24**. The transmitted signal after the phase shift by the variable phase shifter **23** is hereinafter referred to as a cancel signal.

[0035] Thus, the variable attenuator **22** and the variable phase shifter **23** serve as a second generation unit that generates the cancel signal.

[0036] The DA converter **24** converts gain setting data output from the control unit **32** into a gain setting signal in an analog state and supplies the gain setting signal to the variable attenuator **22**. The DA converter converts phase shift amount setting data output from the control unit **32** into a phase shift amount setting signal in an analog state and supplies the phase shift amount setting signal to the variable phase shifter **23**.

[0037] The power combiner **25** power combines a cancel signal output from the variable phase shifter **23** with a received signal output from the output terminal TOA of the antenna duplexer **18**. Accordingly, the power combiner **25** reduces a self-interference signal in the received signal. The power combiner **25** is an example of a suppression unit.

[0038] The quadrature detector **26** performs quadrature detection on a received signal output from the power combiner **25** using the two carrier waves output from the oscillator **11** and the phase shifter **12**. The quadrature detector **26** outputs two-system received baseband signals in an analog state obtained by quadrature detection in parallel.

[0039] The LPF **27** removes an unnecessary frequency component other than a baseband component from each of the two-system received baseband signals output from the quadrature detector **26**.

[0040] The AC coupling amplifier **28** amplifies an AC component corresponding to a response wave while cutting a DC component of each of the two-system received baseband signals that pass through the LPF **27**.

[0041] The AD converter **29** digitizes each of the two-system received baseband signals output from the AC coupling amplifier **28**.

[0042] The LPF **30** removes a harmonic component in each of the two-system received baseband signals output from the quadrature detector **26**.

[0043] The AD converter **31** digitizes each of the two-system received baseband signals output from the LPF **30**.

[0044] The memory **33** stores an information processing program described for information processing to be executed by the CPU **321**. One of the information processing programs stored in the memory **33** is a control program PRA related to a controlling to be described later. The memory **33** stores various types of data required for the CPU **321** to execute various types of information processing. The memory **33** stores various types of data generated or acquired when the CPU **321** executes various types of information processing.

[0045] When communicating with a RFID tag **200**, the CPU **321** outputs the I signal and the Q signal according to a predetermined sequence. The CPU **321** reconstructs data sent from the RFID tag **200** based on two-system received signals digitized by the AD converter **29**. The CPU **321** executes information processing to be described later for controlling the matching adjustment unit **19** or for adjusting a gain in the variable attenuator **22** and a phase shift amount in the variable phase shifter **23**.

[0046] The FPGA **322** executes signal processing programmed in advance to execute various calculations associated with information processing by the CPU **321** at high speed. One of functions of the FPGA **322** is calculating a suppression amount of the self-interference signal due to the cancel signal, based on levels of the two-system received baseband signals digitized by the AD converter **31**. One of the functions of the FPGA **322** is determining a magnitude of a return loss of the antenna **21**, based on levels of the two-system received baseband signals digitized by the AD converter **31**. The return loss of the antenna **21** is a ratio of a magnitude of power supplied to the antenna **21** to a magnitude of reflected power occurring at a feeding point of the antenna **21**.

[0047] FIG. **2** is a block diagram showing a circuit configuration of main parts of the matching adjustment unit **19**.

[0048] The matching adjustment unit **19** includes a plurality of matching circuits **191** and switches **192** and **193**. The switches **192** and **193** insert one of the plurality of matching circuits **191** between the antenna duplexer **18** and the antenna **21** under control of the CPU **321**. Each of the plurality of matching circuits **191** changes an antenna impedance when the matching circuit **191** is inserted between the antenna duplexer **18** and the antenna **21**. Each of the plurality of matching circuits **191** includes an inductor **1911** connected in series between the antenna duplexer **18** and the antenna **21**, and a capacitor **1912** connected in parallel between the antenna duplexer **18** and the antenna **21**. The matching circuits **191** have different combinations of an inductance value of the inductor **1911** and a capacitance value of the capacitor **1912**.

[0049] FIG. **3** is an immittance chart showing adjustment amounts of the plurality of matching circuits **191**.

[0050] As an example, the number of matching circuits **191** is set to 16, and each of the matching circuits **191** is associated with one of 16 antenna impedances VA, VB, . . . , VP shown in FIG. **3**. The antenna impedances VA to VH are determined on a circle corresponding to a return loss of 6 dB. The antenna impedances VI to VP are determined on a circle corresponding to a return loss of 9 dB. The number of matching circuits **191** and characteristics of each matching circuit **191** may be appropriately determined by, for example, a designer of the reading device **100**.

[0051] The matching adjustment unit **19** according to the present embodiment aims to achieve impedance matching so that the return loss is 14 dB or more. That is, an inside of a circle corresponding to the return loss of 14 dB in FIG. **3** is a control target range. If the return loss is 14 dB, a voltage standing wave ratio (VSWR) is approximately 1.5. Since Radio Law stipulates that the voltage standing wave ratio is to be 1.5 or more, it is appropriate to set the control target range for the return loss at 14 dB or more.

[0052] Each matching circuit **191** is configured to match an associated antenna impedance to 50 Ω (an origin in FIG. **3**). Arrows in FIG. **3** indicate matching of the antenna impedance VL to 50 Ω . In the matching circuit **191** for matching the antenna impedance VL to 50 Ω in this manner, an inductance value of the inductor **1911** is LL in FIG. **3**, and a capacitance value of the capacitor **1912** is CL in FIG. **3**.

[0053] Next, an operation of the reading device **100** configured as described above will be described.

[0054] Before describing the operation, the self-interference signal and the cancelling thereof will be described.

[0055] The antenna duplexer **18** is designed so that a transmitted signal input to the input terminal TI is not output from the output terminal TOA. However, in an actual circuit configuration, it is difficult to completely prevent the transmitted signal input to the input terminal TI from leaking from the output terminal TOA. Therefore, a part of the transmitted signal input to the input terminal TI (hereinafter referred to as a leak signal) is output as it is from the output terminal TOA. A part of the transmitted signal output from the input and output terminal TIO of the antenna duplexer **18** is reflected at feeding point of the antenna **21** and transmitted to the antenna duplexer **18** via the feed line **20**. Power of such a reflected signal increases as antenna impedance mismatching increases. Therefore, the power is hereinafter referred to as mismatch reflected power. Such a reflected signal is output from the output terminal TOA by a function of the antenna duplexer **18**. Thus, the signal output from the output terminal TOA of the antenna duplexer **18** includes a composite signal of a leak signal component and a mismatch reflected power component. Such a composite signal is a self-interference signal. Since the leak signal and the reflected signal have an amplitude and a phase, respectively, an amplitude and a phase of the self-interference signal are determined as a magnitude and a direction of a composite vector of a leak signal vector and a reflected signal vector. The impedance of the antenna **21** changes according to an environment around the antenna **21**, such as a proximity state of the RFID tag **200** and another object to the antenna **21**. Therefore, an impedance matching state related to the antenna **21** also varies depending on the environment around the antenna **21**, and the amplitude and phase of the self-interference signal also vary depending on the environment around the antenna **21**.

[0056] The self-interference signal is a signal derived from the transmitted signal. Therefore, the self-interference signal in the received signal can be cancelled by combining the cancel signal generated by changing an amplitude and a phase of a signal branched off from the transmitted signal with the received signal output from the output terminal TOA of the antenna duplexer **18**. In the reading device **100**, the self-interference signal in the received signal is reduced by combining the cancel signal obtained by changing the amplitude and the phase using the variable attenuator **22** and the variable phase shifter **23** with the received signal output from the output terminal TOA of the antenna duplexer **18** by the power combiner **25**. The processing is cancelling, which is hereinafter referred to as self-jammer cancellation (SJC).

[0057] Next, measuring the self-interference signal and the return loss will be described.

[0058] For example, in a state where the cancellation of the self-interference signal is stopped by stopping an output of the cancel signal from the variable phase shifter **23**, if the carrier waves output from the oscillator **11** and the phase shifter **12** are supplied to the input terminal TI of the antenna duplexer **18** as they are, a level of an output signal from the quadrature detector **26** changes according to the self-interference signal. Therefore, the FPGA **322** performs predetermined calculation on the two-system received baseband signals in a digital state output from the AD converter **31**, thereby determining the amplitude and phase of the self-interference signal.

[0059] A ratio of power of the carrier wave supplied to the input terminal TI of the antenna duplexer **18** to power of the leak signal is determined by characteristics of the antenna duplexer **18** and is constant. Therefore, if the power of the carrier wave supplied to the input terminal TI of the antenna duplexer **18** is constant, the power of the leak signal is also constant. Thus, an amplitude and a phase of the output signal from the quadrature detector **26** change according to the reflected signal. That is, there is a positive correlation between changes in amplitude and phase of the output signal from the quadrature detector **26** and the return loss. Therefore, the FPGA **322** determines the return loss based on the changes in the amplitude and phase of the output signal from the quadrature detector **26**.

[0060] FIGS. 4 and 5 are flowcharts of the controlling.

[0061] If it is necessary to read the RFID tag **200**, the CPU **321** executes the controlling shown in FIGS. 4 and 5 based on the control program PRA.

[0062] As ACT **1** in FIG. 4, the CPU **321** starts transmitting a carrier wave. That is, for example, the CPU **321** brings the quadrature modulator **14** into a state of outputting the carrier waves output from the oscillator **11** and the phase shifter **12** as they are.

[0063] In ACT **2**, the CPU **321** acquires the latest determination result of a return loss by the FPGA **322**.

[0064] In ACT **3**, the CPU **321** checks whether impedance matching performed by the matching adjustment unit **19** is good. For example, the CPU **321** checks whether the return loss acquired in ACT **2** is within the control target range. If the return loss is out of the control target range, the CPU **321** determines the result as NO since the matching is not good, and proceeds to ACT **4**.

[0065] At this time, the CPU **321** detects that the impedance matching related to the antenna **21** is in a predetermined mismatching state in which the return loss is not within the target range, and the CPU **321** has a function as a first detection unit.

[0066] In ACT **4**, the CPU **321** checks whether the impedance matching by the matching adjustment unit **19** is abnormal. For example, the CPU **321** checks whether the return loss acquired in ACT **2** is within a predetermined abnormal range. The abnormal range may be, for example, "less than 6 dB" and may be appropriately determined by a designer of the reading device **100** or the like. If the return loss is out of the abnormal range, the CPU **321** determines NO as the matching is not abnormal, and proceeds to ACT **5**.

[0067] In ACT **5**, the CPU **321** checks whether there is a matching circuit **191** that is not selected while looping through ACT **2** to ACT **7** as described later. If the CPU **321** proceeds from ACT **1** to ACT **6** through ACT **2** to ACT **4**, a matching circuit **191** other than the matching circuit **191** currently selected by the switches **192** and **193** is not selected. If there is such a matching circuit **191** that is not selected, the CPU **321** determines YES and proceeds to ACT **6**.

[0068] In ACT **6**, the CPU **321** switches the switches **192** and **193** so as to select the matching circuit **191** that is not selected.

[0069] In ACT **7**, the CPU **321** updates a switching history. For example, the CPU **321** manages the switching history by storing, for example, in the memory **33**, history data representing a list of matching circuits **191** that are selected among the matching circuits **191** in the matching adjustment unit **19**, and updates the history data to add a matching circuit **191** most recently selected in ACT **6** to the list. Alternatively, the CPU **321** manages the switching history by storing, for example, in the memory **33**, history data representing a list of matching circuits **191** that are not selected among the matching circuits **191** in the matching adjustment unit **19**, and updates the history data to exclude a matching circuit **191** most recently selected in ACT **6** from the list.

[0070] Thereafter, the CPU **321** returns to ACT **2**, and repeats the processing in the same manner as described above. Thus, the CPU **321** repeats the processing of checking the return loss while switching the matching circuit **191** until the impedance matching can be determined as good in ACT **3**, the impedance matching can be determined as abnormal in ACT **4**, or it cannot be determined that there is the unselected matching circuit **191** in ACT **5**. When all the matching circuits **191** are selected by repeating the loop, if the CPU **321** does not determine that the impedance matching is good in ACT **3** or that the impedance matching is abnormal in ACT **4**, the CPU **321** determines NO in ACT **5** and proceeds to ACT **8**. If the return loss acquired in ACT **2** falls within the abnormal range, the CPU **321** determines that the impedance matching is abnormal, determines YES in ACT **4**, and proceeds to ACT **8**. If the CPU **321** proceeds to ACT **8** in this manner, the CPU **321** detects that the impedance matching related to the antenna **21** is in a predetermined difficult-to-match state in which it is difficult to form a good matching state by the matching adjustment unit **19**, and thus has a function as a second detection unit.

[0071] In ACT **8**, the CPU **321** stops transmitting the carrier wave.

[0072] In ACT **9**, the CPU **321** executes error processing. For example, as the error processing, the CPU **321** executes processing to inform an operator that an environment is not appropriate for reading. As the error processing, the CPU **321** may execute other processing such as outputting an alarm sound or reproducing a voice message. The CPU **321** may execute a plurality of types of processing as the error processing. An operation such as a display for guidance associated with the error processing is performed by, for example, another information terminal. The reading device **100** may include an appropriate user interface device (not shown), and the user interface device may perform an operation such as display for guidance associated with the error processing. A content of the guidance given by the error processing may be different for a case where YES is determined in ACT **4** and a case where NO is determined in ACT **5**. Thus, the CPU **321** has a function as a processing unit that executes the error processing. Upon completion of the error processing, the CPU **321** ends current control processing.

[0073] For example, if the return loss falls within the target range by impedance matching via the matching circuit **191** selected by the switches **192** and **193** at the start of the controlling or via the matching circuit **191** sequentially selected thereafter, the CPU **321** determines that the matching is good and determines YES in ACT **3**, and proceeds to ACT **10**. By controlling the switches **192** and **193** by the CPU **321** in this manner, the impedance of the antenna **21** is matched so as to eliminate a mismatching state. That is, the CPU **321** functions as a control unit. A function of the CPU **321** as the control unit and the matching adjustment unit **19** cooperate to implement a function of an adjustment unit.

[0074] In ACT **10**, the CPU **321** starts cancellation. That is, the CPU **321** starts outputting a cancel signal from the variable phase shifter **23**. The CPU **321** sets an attenuation amount of the variable attenuator **22** and a phase shift amount of the variable phase shifter **23** such that an amplitude and a phase of the cancel signal are set to predetermined initial values. The initial values of the amplitude and the phase of the cancel signal are determined within a predetermined sweep range. The sweep range is a range including an amplitude and a phase of the self-interference signal that may occur in a standard situation. The sweep range may be appropriately determined by, for example, a designer of the reading device **100**.

[0075] In ACT **11**, the CPU **321** sweeps the amplitude and phase of the cancel signal within the sweep range. That is, for example, the CPU **321** sequentially changes the attenuation amount of the variable attenuator **22** and the phase shift amount of the variable phase shifter **23** within a predetermined range according to the sweep range. Then, the CPU **321** determines the attenuation amount and the phase shift amount when a level of an output signal of the quadrature detector **26** becomes minimum.

[0076] In ACT **12**, the CPU **321** sets operating states of the variable attenuator **22** and the variable phase shifter **23** to minimum points. That is, for example, the CPU **321** sets the amplitude and the phase of the cancel signal to the attenuation amount and the phase shift amount determined in ACT **11**.

[0077] In ACT **13**, the CPU **321** checks whether a state is a specified suppression state. The specified suppression state is a state in which an influence of the self-interference signal is sufficiently reduced. More specifically, as an example, a state in which a level of a component of the self-interference signal remaining in the output signal of the quadrature detector **26** (hereinafter, referred to as a residual signal) is reduced to -20 dB or less compared with that in a non-suppression state is defined as the specified suppression state. That is, the CPU **321** checks a suppression amount measured by the FPGA **322**, and if it is confirmed that the suppression amount is reduced to -20 dB or less from a time of non-suppression, the CPU **321** determines that the state is the specified suppression state. The specified suppression state is preferably determined as a state in which a minimum suppression amount at which the quadrature detector **26** is not saturated due to an influence of the residual signal is obtained. The specified suppression state is appropriately determined by, for example, a designer of the reading device **100**. For example, when the return

loss is acquired last in ACT 2, the CPU 321 acquires the latest determination result related to the level of the output signal of the quadrature detector 26 by the FPGA 322, and uses the latest determination result as a level of the self-interference signal in the non-suppression state.

[0078] If the CPU 321 confirms that the state is not the specified suppression state, the CPU 321 determines NO in ACT 13 and repeats ACT 2 and the subsequent ACTs. That is, since there is a possibility that the antenna impedance matching is not good due to a change in a surrounding environment or the like, adjustment of the impedance matching described above is restarted.

[0079] If the CPU 321 confirms that the state is the specified suppression state, the CPU 321 determines YES in ACT 13 and proceeds to ACT 21 in FIG. 5.

[0080] If the carrier wave transmitted so far reaches the RFID tag 200, the RFID tag 200 starts generating operation power as rectified power based on the carrier wave.

[0081] In ACT 21, the CPU 321 starts reading the RFID tag 200. That is, the CPU 321 calls one RFID tag 200 activated as described above to transmit a response signal, and starts a series of sequences for receiving the response signal (hereinafter, referred to as a reading sequence). The reading sequence may be similar to that performed by another existing reading device, and a detailed description thereof will be omitted. Then, the CPU 321 shifts to a standby state in ACT 22 and ACT 23 while executing the reading sequence.

[0082] In ACT 22, the CPU 321 checks whether the reading of one RFID tag is completed. Then, if the CPU 321 cannot confirm the corresponding event, the CPU 321 determines NO, and proceeds to ACT 23.

[0083] In ACT 23, the CPU 321 confirms whether a state is the specified suppression state is set, for example, in the same manner as in ACT 13 in FIG. 4. If the specified suppression state remains, the CPU 321 determines YES, and returns to ACT 22.

[0084] Thus, in the standby state of ACT 22 and ACT 23, the CPU 321 waits for the reading to be completed or waits for the state to be no longer in the specified suppression state.

[0085] If the reading of one RFID tag 200 is completed, the CPU 321 determines YES in ACT 22 and proceeds to ACT 24.

[0086] In ACT 24, the CPU 321 checks whether there is another RFID tag 200 to be read. If there is the RFID tag 200 to be read, the CPU 321 determines YES and proceeds to ACT 25.

[0087] In ACT 25, the CPU 321 acquires the latest determination result of a return loss by the FPGA 322.

[0088] In ACT 26, the CPU 321 confirms whether the impedance matching by the matching adjustment unit 19 is good in the same manner as in ACT 3 in FIG. 4. If the CPU 321 confirms that the impedance matching is good, the CPU 321 determines YES, and repeats the processing after ACT 21 in the same manner as described above. That is, the CPU 321 attempts to read another RFID tag 200 to be read.

[0089] During execution of the reading sequence, the self-interference signal may change due to a change in the surrounding environment or the like, and the specified suppression state may not be maintained. In particular, if a communication speed is relatively low and a time required to execute the reading sequence (hereinafter, referred to as reading time) becomes long, there is a concern that the self-interference signal may greatly change during the execution of the reading sequence.

[0090] FIG. 6 is a diagram showing an actual measurement example of variations in a phase of a reflected signal over time.

[0091] FIG. 6 is an example of reading from the RFID tags 200 each attached to each of a plurality of articles placed on a shelf while moving the reading device 100. In FIG. 6, a solid line indicates a case where the reading device 100 is moved in a horizontal direction of the shelf, and a broken line indicates a case where the reading device 100 is moved in a vertical direction of the shelf.

[0092] FIG. 7 is a diagram showing an actual measurement example of a change in a cancellation amount due to a phase error.

[0093] It is desirable to secure a cancellation amount of 20 dB at minimum, and it is desirable to

suppress the phase error within 5 degrees, as can be seen in FIG. 7.

[0094] On the other hand, in FIG. 6, if attention is paid to a portion where variations over time are steep, variations in the 5-degree reflection phase are observed in 31 msec from 524 msec to 555 msec. Therefore, if the reading of one RFID tag **200** is not completed within 31 msec, it is conceivable that the cancellation amount becomes insufficient during the reading and a reading error occurs.

[0095] The reading time for the RFID tag **200** depends on performance of the RFID tag **200**. For example, if the RFID tag **200** is in an ultra high frequency (UHF) band conforming to the currently popular ISO/IEC 18000-63/EPCGEN2 standard, the reading time is about 6 msec (40 kbps, 240 bit length), which is within the above-described 31 msec.

[0096] However, if the RFID tag **200** is configured at a lower cost using print electronics technology or the like and requires a reading time of 100 msec or more, reading is not completed within the above-described 31 msec.

[0097] In particular, if the RFID tag **200** to be read is of such a low-speed type, there is a possibility that the specified suppression state cannot be maintained during execution of the reading sequence. Even if the reading time is short, if a steep environmental variation occurs within the reading time, there is a possibility that the specified suppression state cannot be maintained.

[0098] If the CPU **321** cannot confirm that the RFID tag **200** is in the specified suppression state before the reading of one RFID tag **200** is completed, the CPU **321** determines NO in ACT **23** in FIG. 5 and proceeds to ACT **27**.

[0099] In ACT **27**, the CPU **321** stops the reading sequence being executed. Thereafter, the CPU **321** repeats the processing of ACT **5** and thereafter in FIG. 4 in the same manner as described above.

[0100] On the other hand, if the CPU **321** cannot confirm that the impedance matching by the matching adjustment unit **19** is good after the reading of one RFID tag **200** is completed, the CPU **321** determines NO in ACT **26** and proceeds to ACT **28**.

[0101] In ACT **28**, the CPU **321** confirms whether the impedance matching by the matching adjustment unit **19** is abnormal in the same manner as in ACT **4** in FIG. 4. Then, if the CPU **321** cannot confirm that impedance matching is abnormal, the CPU **321** determines NO, and proceeds to ACT **29**.

[0102] In ACT **29**, the CPU **321** resets a switching history. For example, the CPU **321** updates history data so that it can be determined that only the matching circuit **191** selected by the switches **192** and **193** at this time point is selected. Alternatively, the CPU **321** may update the history data so that it can be determined that no matching circuit **191** is selected. Thereafter, the CPU **321** executes the processing after ACT **2** in FIG. 4 in the same manner as described above. Accordingly, the CPU **321** restarts the above-described impedance matching adjustment from the beginning.

[0103] In contrast, if the CPU **321** confirms that the impedance matching by the matching adjustment unit **19** is abnormal in ACT **28** in FIG. 5, the CPU **321** determines YES and proceeds to ACT **30**. If the CPU **321** proceeds to ACT **30** in this manner, the CPU **321** detects that the impedance matching related to the antenna **21** is in a predetermined difficult-to-match state in which it is difficult to form a good matching state by the matching adjustment unit **19**, and thus has a function as the second detection unit.

[0104] In ACT **30**, the CPU **321** stops transmitting the carrier wave.

[0105] In ACT **31**, the CPU **321** executes error processing as in ACT **9** in FIG. 4. At this time, the CPU **321** functions as the processing unit that executes the error processing. Upon completion of the error processing, the CPU **321** ends current control processing.

[0106] If reading of all the RFID tags **200** to be read is completed, the CPU **321** determines NO in ACT **24** in FIG. 5 and proceeds to ACT **32**.

[0107] In ACT **32**, the CPU **321** stops transmitting the carrier wave. Then, the CPU **321** ends current control processing.

[0108] As described above, if the mismatching state of the impedance matching related to the antenna **21** occurs due to a variation in the surrounding environment, the reading device **100** determines the mismatching state based on a change in the reflected signal, and performs the impedance matching by the matching adjustment unit **19** to achieve the matching state. Accordingly, even if the impedance of the antenna **21** varies, the impedance matching is performed to reduce the self-interference signal, and the self-interference signal can be appropriately cancelled by the SJC.

[0109] The reading device **100** can adjust the antenna impedance at high speed by switching the plurality of matching circuits **191** by the switches **192** and **193**.

[0110] If the impedance matching by the matching adjustment unit **19** is not good even after all the matching circuits **191** in the matching adjustment units **19** are selected, that is, if the impedance cannot be adjusted by the matching adjustment unit **19** so as to make the impedance matching good, the reading device **100** may not be able to perform normal reading, so that an operator or the like is notified of the fact by the error processing. Accordingly, the operator or the like can take appropriate measures such as quickly taking measures such as improving the surrounding environment that causes the impedance variation.

[0111] If the impedance matching by the matching adjustment unit **19** is abnormal, the reading device **100** may not be able to perform normal reading, so that an operator or the like is notified of the fact by the error processing. Accordingly, the operator or the like can take appropriate measures such as quickly taking measures such as improving the surrounding environment that causes the impedance variation.

[0112] In the present embodiment, if the return loss is in an abnormal range of, for example, “less than 6 dB”, the CPU **321** determines that the impedance matching by the matching adjustment unit **19** is abnormal. If the return loss is 6 dB, 50% of power supplied to the antenna **21** is reflected at the antenna feeding point. Therefore, in a situation where the return loss is less than 6 dB, mismatch reflected power is large, and the power exceeds a maximum input (maximum rating) of a semiconductor element used in a receiving circuit, and there is a risk that the element is damaged. In some cases, a protection circuit is provided to prevent damage to the semiconductor element, but in such a case, a function of the protection circuit will cause the reading device **100** to not function. Therefore, it is appropriate to perform the error processing in a situation where the return loss is less than 6 dB.

[0113] The embodiment can be modified in various ways as follows.

[0114] The embodiment may be implemented as a communication device that communicates with a wireless tag such as the RFID tag **200** for the purpose different from the purpose of reading data stored in the RFID tag **200** from the RFID tag **200**.

[0115] Each function implemented by the CPU **321** through the information processing can also be implemented in part or in whole by hardware that executes information processing not based on a program, such as a logic circuit. Each of the above functions can be implemented by combining software control with hardware such as the above logic circuit.

[0116] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the exemplary embodiments. These novel embodiments can be implemented in various other forms, and various omissions, substitutions, and modifications can be made without departing from the gist of the disclosure. The embodiments and the modifications thereof are included in the scope and the spirit of the disclosure, and are included in a scope of the invention disclosed in the claims and equivalents thereof.

Claims

- 1.** A communication device for receiving a response wave emitted from a wireless tag based on a received carrier wave, the communication device comprising: an antenna; a first generation component configured to generate a carrier wave; a duplexer configured to receive the carrier wave generated by the first generation component from an input terminal and output the carrier wave to the antenna from an input and output terminal, and output a signal input from the antenna to the input and output terminal from an output terminal; a second generation component configured to generate a cancel signal by changing an amplitude and a phase of the carrier wave generated by the first generation component; a suppression component configured to suppress a self-interference signal in an output signal from the output terminal by using the cancel signal generated by the second generation component; a first detector configured to detect that impedance matching related to the antenna is in a predetermined mismatching state; and an adjustment component configured to adjust, in response to detection of a mismatching state by the first detector, an antenna impedance to eliminate the mismatching state.
- 2.** The communication device according to claim 1, wherein the adjustment component includes a plurality of matching circuits each configured to match different antenna impedances to a predetermined impedance when connected to the antenna, a switch configured to selectively insert the plurality of matching circuits between the input and output terminal and the antenna, and a controller configured to control the switch to select a matching circuit from the plurality of matching circuits for which the mismatching state is no longer detected by the first detector.
- 3.** The communication device according to claim 1, wherein the adjustment component matches an antenna impedance during a period in which the response wave emitted from the wireless tag is not received.
- 4.** The communication device according to claim 1, wherein the second generation component determines an amount of change in amplitude and phase for generating the cancel signal after the antenna impedance is adjusted by the adjustment component.
- 5.** The communication device according to claim 1, further comprising: a second detector configured to detect a predetermined difficult-to-match state in which impedance matching related to the antenna is difficult; and a processor configured to perform predetermined error processing in response to detection of a difficult-to-match state by the second detector.
- 6.** The communication device according to claim 1, wherein the first generation component comprises at least one of an oscillator and a phase shifter.
- 7.** The communication device according to claim 1, further comprising: a variable phase shifter configured to change a phase of a transmitted signal attenuated by a variable attenuator by a phase shift amount corresponding to a phase shift amount setting signal.
- 8.** The communication device according to claim 1, wherein the second generation component comprises at least one of a variable attenuator and a variable phase shifter.
- 9.** A communication method for receiving a response wave emitted from a wireless tag based on a received carrier wave, the communication method comprising: generating a carrier wave; receiving by a duplexer the carrier wave generated by a first generation component from an input terminal and output the carrier wave to an antenna from an input and output terminal, and outputting a signal input from the antenna to the input and output terminal from an output terminal; generating a cancel signal by a second generation component by changing an amplitude and a phase of the carrier wave generated by the first generation component; a suppression component configured to suppressing a self-interference signal in an output signal from the output terminal by using the cancel signal generated by the second generation component; a first detector configured to detecting that impedance matching related to the antenna is in a predetermined mismatching state; and an adjustment component configured to adjusting, in response to detection of a mismatching state, an antenna impedance to eliminate the mismatching state.
- 10.** The communication device according to claim 9, further comprising: matching, by a plurality

of matching circuits, different antenna impedances to a predetermined impedance when connected to the antenna; selectively inserting the plurality of matching circuits between the input and output terminal and the antenna; and selecting a matching circuit from the plurality of matching circuits for which the mismatching state is no longer detected.

11. The communication device according to claim 9, further comprising: matching an antenna impedance during a period in which the response wave emitted from the wireless tag is not received.

12. The communication device according to claim 9, further comprising: determining an amount of change in amplitude and phase for generating the cancel signal after the antenna impedance is adjusted.

13. An RFID reading device for receiving a response wave emitted from a RFID tag based on a received carrier wave, the RFID reading device comprising: an antenna; a first generation component configured to generate a carrier wave; a duplexer configured to receive the carrier wave generated by the first generation component from an input terminal and output the carrier wave to the antenna from an input and output terminal, and output a signal input from the antenna to the input and output terminal from an output terminal; a second generation component configured to generate a cancel signal by changing an amplitude and a phase of the carrier wave generated by the first generation component; a suppression component configured to suppress a self-interference signal in an output signal from the output terminal by using the cancel signal generated by the second generation component; a first detector configured to detect that impedance matching related to the antenna is in a predetermined mismatching state; and an adjustment component configured to adjust, in response to detection of a mismatching state by the first detector, an antenna impedance to eliminate the mismatching state.

14. The RFID reading device according to claim 13, wherein the adjustment component includes a plurality of matching circuits each configured to match different antenna impedances to a predetermined impedance when connected to the antenna, a switch configured to selectively insert the plurality of matching circuits between the input and output terminal and the antenna, and a controller configured to control the switch to select a matching circuit from the plurality of matching circuits for which the mismatching state is no longer detected by the first detector.

15. The RFID reading device according to claim 13, wherein the adjustment component matches an antenna impedance during a period in which the response wave emitted from the RFID tag is not received.

16. The RFID reading device according to claim 13, wherein the second generation component determines an amount of change in amplitude and phase for generating the cancel signal after the antenna impedance is adjusted by the adjustment component.

17. The RFID reading device according to claim 13, further comprising: a second detector configured to detect a predetermined difficult-to-match state in which impedance matching related to the antenna is difficult; and a processor configured to perform predetermined error processing in response to detection of a difficult-to-match state by the second detector.

18. The RFID reading device according to claim 13, wherein the first generation component comprises at least one of an oscillator and a phase shifter.

19. The RFID reading device according to claim 13, further comprising: a variable phase shifter configured to change a phase of a transmitted signal attenuated by a variable attenuator by a phase shift amount corresponding to a phase shift amount setting signal.

20. The RFID reading device according to claim 13, wherein the second generation component comprises at least one of a variable attenuator and a variable phase shifter.
